Updated Return-to-Play Recommendations for Collision Athletes After Cervical Spine Injury: A Modified Delphi Consensus Study With the Cervical Spine Research Society

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BACKGROUND: Previous studies have attempted to establish return-to-play (RTP) guidelines in collision sport athletes after cervical spine injury; however, recommendations have been limited by scant high-quality evidence and basic consensus survey methodologies.
OBJECTIVE: To create relevant clinical statements regarding management in collision sport athletes after cervical spine injury, and establish consensus RTP recommendations.
METHODS: Following the modified Delphi methodology, a 3 round survey study was conducted with spine surgeons from the Cervical Spine Research Society and National Football League team physicians in order to establish consensus guidelines and develop recommendations for cervical spine injury management in collision sport athletes.
RESULTS: Our study showed strong consensus that asymptomatic athletes without increased magnetic resonance imaging (MRI) T2-signal changes following 1-/2- level anterior cervical discectomy and fusion (ACDF) may RTP, but not after 3-level ACDF (84.4%). Although allowed RTP after 1-/2-level ACDF was noted in various scenarios, the decision was contentious. No consensus RTP for collision athletes after 2-level ACDF was noted. Strong consensus was achieved for RTP in asymptomatic athletes without increased signal changes and spinal canal diameter >10 mm (90.5%), as well as those with resolved MRI signal changes and diameter >13 mm (81.3%). No consensus was achieved in RTP for cases with pseudarthrosis following ACDF. Strong consensus supported a screening MRI before sport participation in athletes with a history of cervical spine injury (78.9%).
CONCLUSION: This study provides modified Delphi process consensus statements regarding cervical spine injury management in collision sport athletes from leading experts in spine surgery, sports injuries, and cervical trauma. Future research should aim to elucidate optimal timelines for RTP, as well as focus on prevention of injuries.

KEY WORDS: Cervical spine, Trauma, Collision, Sports, Return to play, Consensus, Guidelines

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Collision sports have an inherent risk of injury, with estimates reporting 9% to 15% of over 12 000 new cases of spine injuries per year in the United States attributable to sports.[1-6] Based on recent estimates by the U.S. Census Bureau and the U.S. Bureau of Labor Statistics, almost 4 million Americans play football on an average day, and about 80% are between 15 and 24 yr old.[7,8] Spine injuries associated with football are common, with studies suggesting between 7% and 10% of injuries in National Football League (NFL)-level athletes involve the head/neck/spine—not including concussions.[3,9-11] Moreover, estimates suggest 35% to 49% of collision sport spine injuries affect the cervical spine, resulting in nerve injuries (46%), muscle injuries (22%), disc injuries (6%), contusions (2%), and fractures (2%), among

ABBREVIATIONS: ACDF, anterior cervical discectomy and fusion; **AEs,** athlete-exposures; **CSRS,** Cervical Spine Research Society; **MRI,** magnetic resonance imaging; **NFL,** National Football League; **RTP,** return-to-play

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others; and a majority affect individuals under 30 yr old.[1,3,10,12-14] Furthermore, recent reports for NFL-level and college-level football athletes suggest axial skeleton in-game injury incidence rates of 3.55/1000 athlete-exposures (AEs) and 148.26/100 000 AEs, respectively.[4,10]

Over the years, recommendations have been published to direct return-to-play (RTP) decisions in collision sports with respect to a variety of injuries to the cervical spine. Given the complexity and functional consequences of these injuries, prospective, randomized, controlled trials are not plausible; as a result, most available guidelines for RTP derive from systematic reviews, expert opinion, meta-analyses, and case series or reports.[15-28] However, consensus RTP recommendations have yet to be universally established.[29] The purpose of this study was to utilize a modified Delphi process with the Cervical Spine Research Society (CSRS) to formulate actionable statements and establish updated consensus RTP recommendations for providers caring for collision athletes with cervical spine injuries.

METHODS

A modified Delphi method study was conducted using CSRS members and NFL team physicians to build consensus on RTP recommendations after cervical spine injury in collision sports athletes.[30,31] A first-round anonymous online survey composed of 14 clinical scenarios and 38 related questions involving cervical spine injuries in football players was developed based on literature review and expert opinion (**Supplemental Digital Content 1**).[1,30,31] In November 2019, the survey was sent to all CSRS members. Results were reviewed by CSRS members and NFL team physicians, and thirteen consensus statements were developed.[31] These statements were administered to attendees at the CSRS annual meeting in November 2019 (second-round). Voting abstentions were not counted for consensus tallying purposes.[32,33] Results of this second survey were used to generate consensus recommendations based on the following levels: simple majority (50.1%-59%, no consensus), super majority (60%-70%, weak consensus), super majority (71%-94%, strong consensus), and unanimous (95%-100%, strongest consensus). To address concerns with RTP for professional athletes after 2-level anterior cervical discectomy and fusion (ACDF) based on second-round data, a third round online anonymous survey was sent to a focused group of NFL-affiliated spine surgeons. A Fisher exact test was performed to compare data between professional and nonprofessional sports affiliated physicians where appropriate, with significance established at $P < .05$. This clinician survey study was exempt from Institutional Review Board review.

RESULTS

Initial Cervical Trauma Survey

First round responses were received from 48 expert spine surgeon members of the CSRS: 87.5% were orthopaedic surgeons (n = 42) and 12.5% were neurosurgeons (n = 6). Table 1 shows characteristics of the initial expert panel. For details of the results of the initial survey, see **Supplemental Digital Content 2**.

TABLE 1. Expert Panel Demographics for First Round Questionnaire

Initial expert panel demographics (n = 48)	Number of experts
Specialty	
Orthopaedic spine surgery	42
Neurosurgery	6
Years in practice	
<5 yr	3
5-10 yr	5
10-20 yr	19
>20 yr	21
Team physician (NCAA or NFL)	
Yes	12
No	36

NCAA = National Collegiate Athletic Association; NFL = National Football League.

RTP Consensus Recommendations

Upon analysis of initial survey results, 13 clinical recommendations were developed and voted upon by 95 attendees of the 2019 CSRS Annual Meeting, with an average engagement of 71%. Of these participants, 64/95 identified as orthopaedic spine surgeons and 3/95 identified as "other", with 28/95 participants not identifying their subspecialty. Additionally, 17/95 participants identified as actively caring for professional collision athletes, and 63/95 identified as having involvement in the care of competitive collision athletes. From the 13 recommendations surveyed, 8 statements achieved strong consensus, 3 achieved weak consensus, and 2 achieved no consensus. No significant difference in responses was noted between professional and nonprofessional sports team-affiliated physicians.

RTP Recommendations in Collision Sport Athletes Based on Severity of Cervical Canal Stenosis

Regarding collision athletes after an initial episode of transient paralysis from a cervical spine injury, 90.5% (strong consensus) of participants agreed that RTP was recommended once symptoms have subsided, there are no T2 magnetic resonance imaging (MRI) changes and spinal canal diameter is >10 mm; if canal diameter is in the absolute stenosis range, RTP should be considered on an individual basis. Additionally, for athletes in a similar case but with initial changes on T2 sequence MRI, almost 81.3% (strong consensus) of participants noted that RTP was allowed once MRI changes resolved only if spinal canal diameter is >13 mm; for those in the relative stenosis range RTP should be approached on a case-by-case basis, and those with absolute stenosis should not be allowed to RTP. Furthermore, 68.3% (weak consensus) of participants noted that for athletes that have persistent MRI changes, individual evaluation for RTP can be considered only if spinal canal diameter is >13 mm, otherwise RTP should be contraindicated. Of note, only one statement achieved a simple majority without consensus for a disagreement, with 50.7% of surgeons (38.5% of those affiliated

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TABLE 2. RTP Guidelines Based on Severity of Cervical Stenosis

Statement	All physicians % (n)	Level of recommendation	% Agreement of professional sports team-affiliated physicians[a] % (n)	*P* value[b]
RTP guidelines in collision athletes based on severity of cervical stenosis				
Following an episode of transient paralysis, asymptomatic athletes with NO T2-signal change and a spinal canal diameter >10 mm are allowed to RTP, but those with a canal diameter <10 mm should be taken on a case-by-case basis.	**Agree: 90.5 (57)** Disagree: 9.5 (6)	Super majority, strong consensus	85.7 (12/14) Strong consensus	.64
Following an episode of transient paralysis, asymptomatic athletes with RESOLVED T2-signal changes and a spinal canal diameter >13 mm are allowed to RTP; those with a canal diameter between 10 and 13 mm should be considered on a case-by-case basis, and those with a canal diameter <10 mm should not RTP.	**Agree: 81.25 (52)** Disagree: 18.75 (12)	Super majority, strong consensus	66.7 (10/15) Weak consensus	.10
Following an episode of transient paralysis, for asymptomatic athletes with CONTINUED T2-signal change and a spinal canal diameter >13 mm RTP should be taken on a case-by-case basis, but those with a canal diameter <13 mm should not RTP.	**Agree: 68.3 (41)** Disagree: 31.7 (19)	Super majority, weak consensus	92.3 (12/13) Strong consensus	.15
Following an injury, for asymptomatic athletes with no history of transient paralysis with CONTINUED T2-signal changes and a spinal canal diameter >13 mm RTP should be taken on a case-by-case basis, but those with canal diameters <13 mm should not RTP.	Agree: 49.3 (33) **Disagree: 50.7 (34)**	Simple majority, no consensus	38.5 (5/13) Weak consensus disagreement	.75

[a] A total of 17 physicians in the survey reported being affiliated with a professional collision/contact sports team.
[b] Fisher exact test comparing statement agreement between physicians affiliated and not affiliated with professional sports teams.
Bold highlights the response with the highest percentage.

with a professional team) disagreeing that RTP in asymptomatic athletes without evidence of cervical stenosis following a cervical spine injury and no history of transient paralysis but persistent MRI T2-signal changes should be approached on a case-by-case basis, but if there is any evidence of cervical stenosis, RTP should not be allowed (Table 2).

RTP Recommendations in Collision Sport Athletes After Cervical Spine Trauma and/or Surgery

For asymptomatic collision athletes after a cervical spine injury, 84.4% (strong consensus) of surgeons noted that solidly fused 1-/2-level ACDF patients could RTP when there are no MRI T2-signal changes; however, 3-level ACDF patients should not RTP, even without increased MRI changes. Similarly, athletes after a 1-level ACDF with continued MRI changes and a solid fusion (81.9%, strong consensus) or with pseudarthrosis (59.5%, simple majority without consensus) can RTP on a case-by-case basis, whereas multi-level ACDF patients should not RTP. Interestingly, 62.5% (weak consensus) of professional sports team-affiliated responders would consider RTP on a case-by-case basis for athletes with a 1-level ACDF and pseudarthrosis. For athletes with cervical spine fractures, 98.6% (strong consensus) would allow RTP as long as there is no instability or increased T2-signal changes. Remarkably, 100% of professional team-affiliated physicians would allow these patients to RTP. Finally, for athletes with solid fusion after a cervical spine fracture, RTP can be allowed if there are no MRI changes following a 1-/2-level ACDF or 1-level posterior cervical fusion (68.6%, weak consensus). Professional team-affiliated surgeons reached a strong consensus (75%) on this observation. Of note, for athletes with resolved transient paralysis and asymptomatic, 72.9% (strong consensus) of surgeons permitted RTP after a 1-/2-level ACDF if there were no T2-signal changes postoperatively, but case-by-case consideration was recommended after a corpectomy or posterior surgery (Table 3).

Third-round survey results for clarification of RTP in collision athletes after 2-level ACDF showed that the decision is controversial with only 50% (4/8) of responders allowing return to sport (Table 4).

Additional Considerations for RTP in Collision Sport Athletes

In terms of imaging in collision athletes after a cervical spine injury, 78.9% (strong consensus) of participants would obtain a screening MRI prior to participation clearance in athletes with a prior history of cervical spine pathology—excluding stingers. This same finding noted weak consensus among professional team-affiliated surgeons (69.2%). Moreover, for stingers, 69.9% (weak consensus) of surgeons would not obtain an MRI if symptoms resolved quickly after a first event, but any subsequent event would require MRI. Finally, RTP would be allowed by 84.5% (strong consensus) of participants if stinger symptoms resolved

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TABLE 3. RTP Guidelines After Cervical Trauma/Surgery

Statement	All physicians % (n)	Level of recommendation	% Agreement of professional sports team-affiliated physicians[a] % (n/responders)	*P* value[b]
RTP guidelines in collision athletes after cervical trauma and/or surgery				
Asymptomatic athletes with NO T2-signal change after a solid 1-/2-level ACDF are allowed to RTP, but a 3-level ACDF should not RTP.	**Agree: 84.4 (65)** Disagree: 15.6 (12)	Super majority, strong consensus	87.5 (14/16) Strong consensus	.52
Asymptomatic athletes with CONTINUED T2-signal change after a solid 2-/3-level ACDF should not RTP, but a 1-level ACDF should be taken on a case-by-case basis.	**Agree: 81.9 (59)** Disagree: 18.1 (13)	Super majority, strong consensus	86.7 (13/15) Strong consensus	.58
Athletes with pseudarthrosis following a 2-/3-level ACDF should not RTP, but pseudarthrosis of a 1-level ACDF should be taken on a case-by-case basis.	**Agree: 59.5 (44)** Disagree: 40.5 (30)	Simple majority, no consensus	62.5 (10/16) Weak consensus	.55
Asymptomatic athletes with a solid fusion after a compression fracture, burst fracture, or facet fracture with no instability and no T2-signal change are allowed to RTP.	**Agree: 98.6 (72)** Disagree: 1.4 (1)	Super majority, strong consensus	100 (16/16) Unanimous consensus	1.00
Asymptomatic athletes with a solid fusion after a cervical spine fracture and NO T2-signal change following a 1-/2-level ACDF or 1-level PCF are allowed to RTP.	**Agree: 68.6 (48)** Disagree: 31.4 (22)	Super majority, weak consensus	75 (12/16) Strong consensus	.54
Following an episode of transient paralysis, asymptomatic athletes with NO T2-signal change following a 1-/2-level ACDF are allowed to RTP, but following a corpectomy or posterior cervical surgery RTP should be taken on a case-by-case basis.	**Agree: 72.9 (43)** Disagree: 27.1 (16)	Super majority, strong consensus	64.3 (9/14) Weak consensus	.51

[a] A total of 17 physicians in the survey reported being affiliated with a professional collision/contact sports team.
[b] Fisher exact test comparing statement agreement between physicians affiliated and not affiliated with professional sports teams.
Bold highlights the response with the highest percentage.

within 5 min, but any episode lasting >5 min would require further evaluation (Table 5).

DISCUSSION

Universally accepted RTP recommendations in collision athletes after cervical spine injury have been difficult to develop. As recently as 2017, Nagoshi and colleagues[24] conducted a systematic review of RTP and cervical spine injuries. The authors noted a lack of high-level evidence throughout, with high risk of bias across the 16 studies meeting inclusion criteria. As a result, the report was unable to determine evidence-based RTP guidelines and resorted to expert opinion for controversial recommendations.[24] France and colleagues[34] in 2016 performed a survey of spine trauma surgeons treating cervical spine injuries to provide RTP recommendations based on a limited consensus expert opinion survey of spine trauma surgeons. Their study polled 25 surgeons from the Spine Trauma Study Group using 10 clinical scenarios of athletic cervical spine injuries, asking them to identify RTP recommendations after symptom recovery. Although the study did find consensus RTP in high-contact sports in the setting of cervical cord neuropraxia with full symptom resolution and no spinal canal stenosis, variability in clinical recommendations was observed in more complex scenarios with persistent symptoms, spinal canal stenosis or previous history of cervical fusion.[34] Moreover, the study was limited by the small number of experts surveyed, as well as the use of a simple survey methodology. Due to the lack of research on the topic and weak consensus achieved by previous studies, our team sought to develop improved guidelines. Our current consensus study was based on a proven process regularly used in the medical literature for developing and measuring consensus.[30]

RTP Recommendations in Collision Sport Athletes Based on Severity of Cervical Canal Stenosis

The presence of cervical spine stenosis has been a key factor considered in RTP decisions for collision sport athletes; however, the definition of cervical stenosis and how it relates to cervical spine injury has been a contentious topic.[25] The literature defines absolute cervical stenosis as an average cervical spine sagittal

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TABLE 4. Third-Round 1-/2-Level ACDF Clarification Survey

Question	All physicians % (n)	Level of recommendation
Third-round 2-level ACDF clarification survey		
Do you take care of American professional football players?	**Yes: 100.0 (8)** No: 0.0 (0)	N/A
Would you allow an asymptomatic player with full range of motion, no neurologic deficits, and no T2-signal changes on MRI, with a solid 1-level ACDF to return to play American football?	**Agree: 100.0 (8)** Disagree: 0.0 (0)	Unanimous, strongest consensus
Would you allow an asymptomatic player with full range of motion, no neurologic deficits, and no T2-signal changes on MRI, with a solid 2-level ACDF to return to play American football?	Agree: 50.0 (4) Disagree: 50.0 (4)	No consensus

RTP = return-to-play; MRI = magnetic resonance imaging; ACDF = anterior cervical discectomy and fusion.
Bold highlights the response with the highest percentage.

TABLE 5. Additional Consideration in RTP Guidelines

Statement	All physicians % (n)	Level of recommendation	% Agreement of professional sports team-affiliated physicians[a] % (n)	*P* value[b]
Additional considerations for RTP guidelines in collision athletes				
Athletes with prior nonoperative or operative treatment for cervical spinal pathology (with the exception of a stinger) should undergo a screening MRI prior to playing a competitive collision/contact sports.	**Agree: 78.9 (45)** Disagree: 21.1 (12)	Super majority, strong consensus	69.2 (9/13) Weak consensus	.45
Athletes who experience their first stinger with rapid resolution of symptoms do not require an MRI, however an MRI should be obtained after a second stinger.	**Agree: 69.9 (41)** Disagree: 30.5 (18)	Super majority, weak consensus	75 (9/12) Strong consensus	.50
Athletes who are asymptomatic for <5 min following a stinger are allowed to RTP, but for those with symptoms lasting >5 min RTP should be taken on a case-by-case basis.	**Agree: 84.5 (49)** Disagree: 15.5 (9)	Super majority, strong consensus	92.9 (13/14) Strong consensus	.41

[a] A total of 17 physicians in the survey reported being affiliated with a professional collision/contact sports team.
[b] Fisher exact test comparing statement agreement between physicians affiliated and not affiliated with professional sports teams.
Bold highlights the response with the highest percentage.

diameter <10 mm; whereas relative stenosis is defined as a canal diameter <13 mm.[21,35-37] Torg and colleagues developed the Torg-Pavlov ratio to evaluate the relationship between cervical spine stenosis and cervical spine injury. The measurement is defined as the ratio of the cervical canal diameter over the cervical vertebral body width, with a ratio of 0.8 or less indicating the presence of stenosis.[16,21,38] In addition, the implication of T2-signal hyperintensity changes on MRI after cervical spine injury in athletes is an important factor to consider in RTP decisions. T2 sequence changes generally indicate a traumatic event leading to edema and inflammation, with persistent or chronic signal changes linked to neurodegeneration and demyelination.[39-43] Tempel and colleagues[39] recently investigated these imaging changes, concluding that if athletes are asymptomatic with normal neurologic exams and no spine instability, RTP should not be contraindicated. Further, they noted that clearing an athlete for sports-related activity should not depend upon resolution of T2 hyperintensity changes. Based on these parameters and observations, once a collision sport athlete has experienced an episode of cervical spine injury in the form of transient paralysis or neuropraxia, the results of our consensus survey recommend that asymptomatic athletes without T2-signal changes on cervical MRI and without evidence of absolute cervical stenosis, or those with resolved T2-signal changes and without any evidence of cervical spine stenosis, are allowed to RTP. However, asymptomatic athletes with no MRI changes and with evidence of

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absolute cervical stenosis, those with resolved MRI changes and relative stenosis, or those with continued T2-signal changes and without evidence of any cervical stenosis, are allowed to RTP based on a case-by-case evaluation. Athletes with persistent T2-sequence changes and any form of cervical stenosis, or those with resolved MRI changes and absolute cervical stenosis, should not RTP. These recommendations have also been maintained by other recent guidelines, including Torg et al findings noting that cervical stenosis in stable spines do not predispose athletes to catastrophic injury and RTP should not be obstructed.[16,24,34,38] Nevertheless, clinical caution should always be exercised with close observation and counseling.

RTP Recommendations in Collision Sport Athletes After Cervical Spine Trauma and/or Surgery

Several previous studies reporting outcomes after cervical spine surgery in professional athletes exist, with most suggesting a majority of athletes will RTP within a year.[20,22,44-46] However, these studies are mostly limited to 1-level ACDFs, and data regarding 2 + fusion levels or posterior surgery is scant. Therefore, most clinicians note that decisions based on any guidelines should be made cautiously.[24,34,44] Nevertheless, our consensus study determined that asymptomatic athletes without T2-signal changes on MRI following a 1- or 2-level ACDF could be allowed to RTP; however, corpectomy or posterior cervical surgery RTP decisions should be made on a case-by-case basis, and 3-level ACDF athletes should not RTP. Interestingly, when considering only professional team-affiliated physicians, the observed overall strong consensus for 1-/2-level ACDF vs corpectomy/posterior surgery dropped to weak consensus (72.9% to 64.3%), reflecting the caution to fully endorse guidelines in high-level collision athletes without sufficient evidence available. Moreover, our panel agreed that 1-level ACDF athletes with continued T2-signal changes are not completely contraindicated to RTP, and that their status be evaluated on a case-by-case basis. This was not the case for 2 + level ACDF, where our panel agreed no RTP is indicated when T2-signal changes persist. As previously mentioned, multi-level fusions or surgical approaches other than ACDF tend to be controversial, and clinical decisions for RTP should be carefully addressed on a case-by-case basis.[16,18,22]

Previous reports have noted that although cervical fractures in collision athletes are rare, the inherent danger of the involved sports renders them difficult to consider in the setting of RTP, and they often carry the longest recovery times.[1,18,26] Like other guidelines, our panel agreed that athletes with healed fractures without instability or MRI changes should be allowed to RTP, even in the setting of surgical intervention with a 1-/2-level ACDF or 1-level PCF as long as there is solid fusion and no persistent MRI changes.[1,17,18,26,34] The literature does endorse that pain-free range of motion, evidence of stability on dynamic imaging, a normal neurologic exam, and healed fusions are necessary components of the RTP decision process.[1,17,18,22,26] Remarkably, there was unanimous consensus among participating professional team-affiliated physicians regarding RTP in athletes with a solid fusion for cervical spine fractures and no associated dynamic instability or persistent MRI changes.

Albeit consensus was reached regarding 1-/2-level ACDF in various scenarios, various responders voiced concern for 2-level ACDF cases in professional athletes. The third-round questionnaire was done in an attempt to appease these concerns utilizing an anonymous survey of 8 surgeons with confirmed involvement in the care of NFL athletes. The results corroborated the controversial nature of RTP decisions in these instances, as no consensus was reached regarding RTP after 2-level ACDF. Our results showed 50% of responders would not allow RTP even in solidly fused 2-level ACDF asymptomatic athletes with full range of motion and no neurologic deficits or MRI T2-signal changes. Given the risks of future injury and long-term consequences, any RTP decisions in this particular patient population need to be carefully considered jointly by the treating physician and the player on a case-by-case basis.[44]

Additional Considerations for RTP in Collision Sport Athletes

Previous studies have addressed the impact a past musculoskeletal medical history has on the career of professional-level football athletes.[47,48] Wang and colleagues studied NFL Scouting Combine participants, and determined that athletes with a history of cervical spine injury have significantly shorter careers and play in fewer games compared to controls.[48] Additionally, among NFL athletes, Schroeder et al determined that professional career outcomes depend on the type of cervical spine pathology an athlete may have, and concluded that each player with a history of cervical spine disease should have individualized evaluations to ensure productive and healthy NFL careers.[47] Supporting these observations, our panel agreed with strong consensus that athletes with a history of cervical spine pathology warrant a screening MRI prior to collision sports participation. Interestingly, professional sport affiliated participants also agreed with the statement, albeit with weak consensus.

For milder forms of cervical spine-associated neuropraxias, such as burners or stingers, timing of advanced imaging is debatable. Stingers are an episodic unilateral peripheral nervous system dysfunction, secondary to a compressive-type or traction-type nerve root or brachial plexus traumatic event.[1,25,26] Stingers are associated with dysesthesias, including burning, stinging, weakness and pain, and most episodes last minutes to hours before full symptom resolution [1,18,25,26] Almost all guidelines allow RTP as soon as symptoms revolve, and there is a documented normal neurologic examination with full active range of motion.[1,16,18,26] Most of these guidelines endorse advanced imaging after 3 or more stinger episodes or persistent symptoms, with RTP contraindicated until further work-up is performed to rule out a possible primary cervical spine abnormality.[1,16,18,25,26] Our current consensus agreed with advanced imaging requirements after a second stinger event, as long as the initial event had rapid symptom resolution. Furthermore, the panel agreed that symptomatic stingers for less than 5 min allow RTP for a collision

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athlete, whereas episodes lasting greater than 5 min warrant further consideration.

Limitations

This study was not without limitations. First, dependence on a panel of experts for clinical recommendations is suboptimal, yet in the absence of strong literature evidence, it is a suitable foundation for clinical practice recommendations. Second, another limitation of this study was the low rate of neurosurgeon participation. An attempt was made to include a breath of subspecialty participants during the 2019 CSRS annual meeting; however, not all participants identified their subspecialty field. Third, the survey process did not allow our team to account for all possible types of cervical spine injuries, surgeries or related considerations in collision athletes. Finally, even though we used a widely accepted process of consensus measurement, our results should not be interpreted as scientific fact, as they are only a measurement of expert opinion. This study does build upon previous guidelines and provide greater consensus strength given the modified Delphi process, as well as the larger number of participants.

CONCLUSION

While cervical spine injuries are rarely experienced by collision sport athletes, their long-term, life-altering effects warrant adequate clinical evaluation and effective nonoperative vs operative management. Though previous studies have provided limited RTP guidelines, the present study utilizes a systematic approach to establish greater consensus statement recommendations for the clinical RTP management of collision sport athletes with cervical spine injury. The results of this study provide clinical recommendations agreed upon by a large panel of leading spine surgeons. Notwithstanding, RTP after 2-level ACDF in this cohort of patients remains controversial. High-quality studies are likely to be impractical given the nature of the injury and situation; however, future studies are required to elucidate the most effective, up-to-date treatment plan and optimal timing for RTP following cervical spine injury.

Disclosures

The authors have no personal, financial, or institutional interest in any of the drugs, materials, or devices described in this article.

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Supplemental Digital Content 1. Initial Questionnaire.
Supplemental Digital Content 2. Supplemental Tables.
Supplemental Table 1. RTP based on levels fused and MRI findings; RTP = return-to-play; MRI = magnetic resonance imaging; ACDF = Anterior Cervical Discectomy and Fusion.
Supplemental Table 2. RTP based on degree of stenosis adjacent to a solid fusion; RTP = return-to-play.
Supplemental Table 3. RTP based on levels fused and pseudarthrosis; RTP = return-to-play; ACDF = Anterior Cervical Discectomy and Fusion.
Supplemental Table 4. RTP Based on fracture type and instability; RTP = return-to-play.
Supplemental Table 5. RTP Based on surgical procedures; RTP = return-to-play; ACDF = Anterior Cervical Discectomy and Fusion.
Supplemental Table 6. RTP in an asymptomatic athlete without history of transient paralysis based on degree of stenosis and MRI findings; RTP = return-to-play; MRI = magnetic resonance imaging.
Supplemental Table 7. RTP in an asymptomatic athlete without history of transient paralysis based on MRI findings and surgical procedures; RTP = return-to-play; MRI = magnetic resonance imaging; ACDF = Anterior Cervical Discectomy and Fusion.
Supplemental Table 8. RTP in an asymptomatic athlete with history of transient paralysis based on degree of stenosis and MRI findings; RTP = return-to-play; MRI = magnetic resonance imaging.
Supplemental Table 9. RTP after an episode of transient paresis based on MRI Findings and surgical procedure; RTP = return-to-play; MRI = magnetic resonance imaging; ACDF = Anterior Cervical Discectomy and Fusion.
Supplemental Table 10. RTP After a Burner/Stinger Based on Symptoms; RTP = return-to-play; MRI = magnetic resonance imaging.
Supplemental Table 11. Recommendations for obtaining an MRI based on previous medical history; MRI = magnetic resonance imaging.

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